

## ROB521 Lab 4

### Task 1.

– present a picture of the AMCL rviz window with a converged filter.

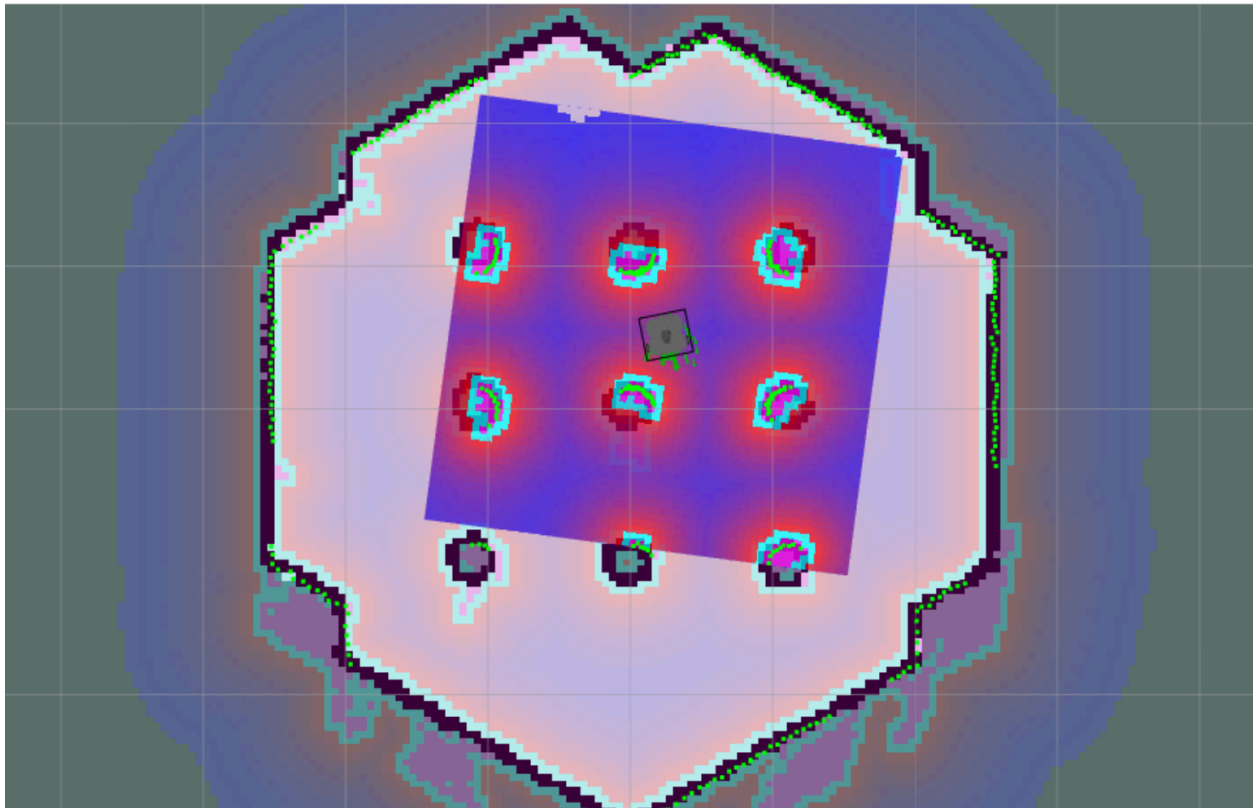


Figure 1: AMCL rviz window with a converged filter.

– what was the minimum number of particles for successful localization?

Are your results repeatable (i.e if you give the robot a new initial pose estimate does the particle filter converge)?

Initial:

Min particles: 500, max particles: 3000. Converges

Experiments:

Min particles: 50, max particles: 300. Converges

Min particles: 25, max particles: 150. Converges after driving for a bit

Min particles: 10, max particles: 60. Unable to converge

The minimum number of particles for successful localization is between 10-25 for the minimal number of particles and 60-150 for the maximum number of particles. Results are repeatable because each experiment was run twice, each time with a slightly different initial pose estimate and a different driver.

– what were the maximum odom\_alpha parameters before localization failed?

Experiments:

Odom value = 5, does not converge

Odom value = 2 does not converge

Odom value = 1.5, converges

Around odom values of 1.5 to 2 is where localization failed.

## Task 2.

– present a picture of your Lab 3 map and your new map. Figure 2 is a solution map for Lab 3. You may use this map if your group has not completed Lab 3.

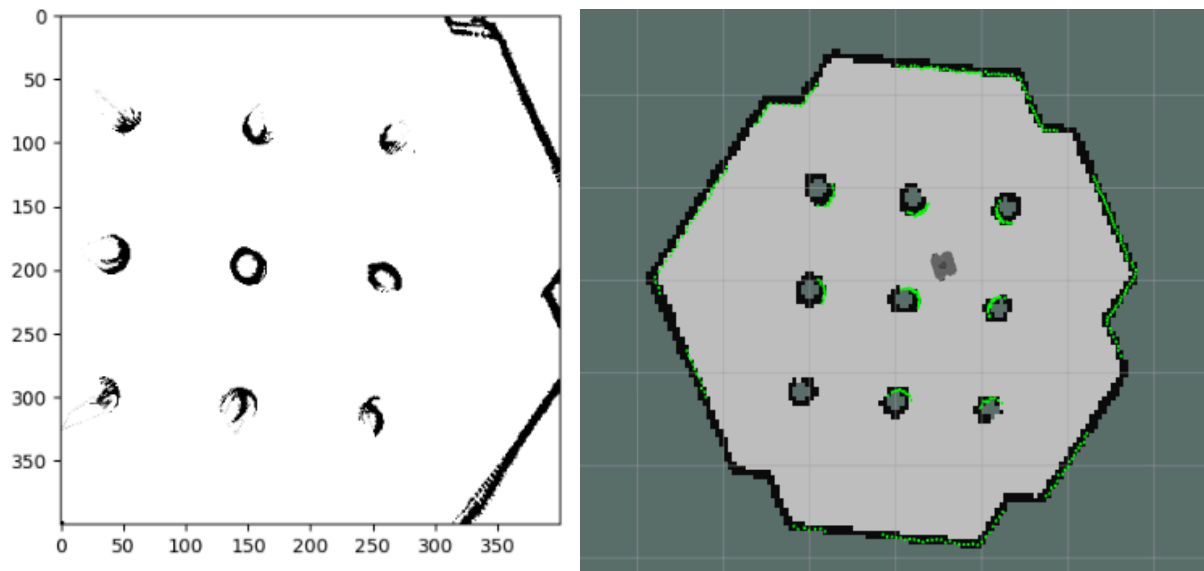


Figure 2: (Left) Solution map for Lab 3. (Right) Solution map using GMapping as required.

– discuss how the localization component of gmapping improved your new map compared to the map from Lab 3.

In the map produced during Lab 3 using only wheel odometry, noticeable blurring and slight misalignments appear along walls and pillars. These distortions arise because errors in wheel odometry, such as slippage, accumulate over time without any correction mechanism, causing the map to become increasingly warped. In contrast, the new map created with Gmapping is much clearer. Walls appear crisp, and pillars and other obstacles line up accurately. This increased clarity reflects Gmapping's ability to correct for drift by using the laser scanner.

Gmapping leverages a Rao-Blackwellized Particle Filter to keep track of multiple possible robot poses. At each time step, it aligns the incoming laser scan with the existing map, ensuring that any deviations from the real path—caused by odometry noise—are corrected by the best-matching hypothesis. Thus, rather than letting wheel odometry errors grow unchecked, Gmapping actively adjusts the robot's pose estimate based on sensor data. When the robot revisits or overlaps previously mapped areas, scan matching further refines alignment in the map, effectively snapping the robot's trajectory back in place if odometry has drifted. This

process reduces the overlapping and “blurry” effect present in maps that rely solely on wheel odometry.

Overall, the main improvement stems from Gmapping’s combination of laser-based localization and map building, which keeps the environment representation both up-to-date and geometrically consistent. This is why the Gmapping map appears much sharper and more accurate than the map created using wheel odometry alone. There’s notably less noise and overlapping on the gmapping map. The map was generated with less distance travelled and it seems more robust at higher speeds and rotations.

### Task 3:

– *present your map and the map provided for Lab 2.*

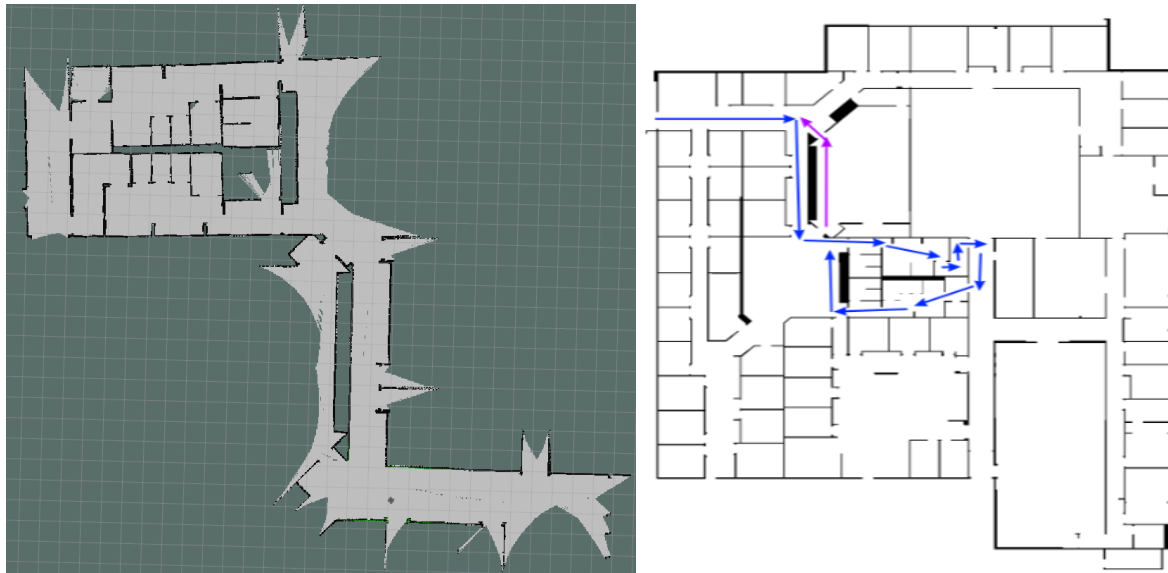


Figure 3: (Left) Our map and (Right) Map provided for Lab 2

– *compare your map to the corresponding part of the map from Lab 2.*

When focusing on just the overlapping sections, the general corridor layout and room positioning match reasonably well between our map and the Lab 2 map. Certain rooms and hallways align closely, showing that Gmapping did a decent job of localizing the robot and updating the map in real time.

However, the Lab 2 map is noticeably cleaner, suggesting that our map experiences some drift or missed loop closures. Edges of corridors occasionally appear jagged or offset, and some doorways are not perfectly rectangular. These differences illustrate the challenges of incremental mapping versus a more thoroughly tuned or pre-processed map like the official Lab 2 reference.

*– describe any mapping issues you encountered and discuss their cause.*

One recurring challenge was drift and misalignment; without effective loop closure, the robot's estimated position gradually diverged from reality, causing warped or disconnected corridors in the map. This drift was exacerbated by poor loop closure scenarios, where Gmapping sometimes failed to recognize revisited areas—especially if we returned at a different angle or moved too quickly—thus missing opportunities to correct the accumulated error. Additionally, sensor noise and motion errors came into play whenever the robot turned too sharply or traveled at high speed, reducing LIDAR scan quality and undermining the reliability of scan matching.

We also discovered the importance of parameter sensitivity in Gmapping's particle filter. Initially, default settings did not cope well with the Willow Garage environment. Adjusting parameters such as maxUrange (to handle longer corridors) and increasing the number of particles helped the filter remain more robust over extended mapping sessions. Likewise, reducing the motion noise parameters (srr, srt, str, stt) improved the odometry model, thus raising the chances of successful loop closure. In short, a combination of environmental factors and suboptimal parameter tuning caused the bulk of our mapping issues. The figure 3 shows results with maxUrange set to 3.5, srr = 0.05, srt = 0.1, str = 0.05, stt = 0.1, particles = 1000. In this run, we also explored side rooms, so loop closure was effective. At times, we did accidentally rotate too fast, which could accumulate odometry drift before loop closure correction.

*– Describe any issues that you encountered when you attempted to close a loop. If the loop closed perfectly, describe issues that you expect you could have encountered.*

On the blue path, which was expected to succeed, the robot did successfully close loops when we moved slowly and kept high overlap between scans. However, if we approached the same area at too different an angle or if there were geometric occlusions, Gmapping occasionally failed to detect we were revisiting a known location. This intermittent loss of loop closure meant that misalignments in the map persisted longer before being corrected. Another factor was the robot's speed: driving too fast magnified motion blur and sensor noise, making it difficult for the SLAM algorithm to match scans accurately.

The purple path typically led to loop-closure failure due to longer stretches of corridor without revisiting previously scanned areas, thereby offering fewer chances to correct drift. Low scan overlap and increased uncertainty over distance (wall looks same along corridor) further strained the filter, eventually causing divergence in the robot's pose estimate. As a result, certain sections of the map became distorted/disconnected. Even if a loop had closed perfectly, one might still encounter abrupt map realignments, overlapping walls, or momentary reversals in robot pose—common side effects when large accumulated errors suddenly get corrected during loop closure. To explain the aforementioned phenomenon, consider the following. Every particle has its own occupancy grid. Map bending and shaking is an illusion, we are showing a copy of the map for the most likely particle (lab), and that particle can change, causing abrupt map changes. Particles that can explain measurements well will become more likely in the loop closure. Note that this means when we close the loop there has to be at least one particle that is close to the real trajectory. If the loop is big enough that particles spread out enough around the

loop that the real answer lies in the gap between particles, then we don't recover and particle filter fails. To avoid this, we used a lot of particles.

#### Task 4

– *present your map of Myhal.*

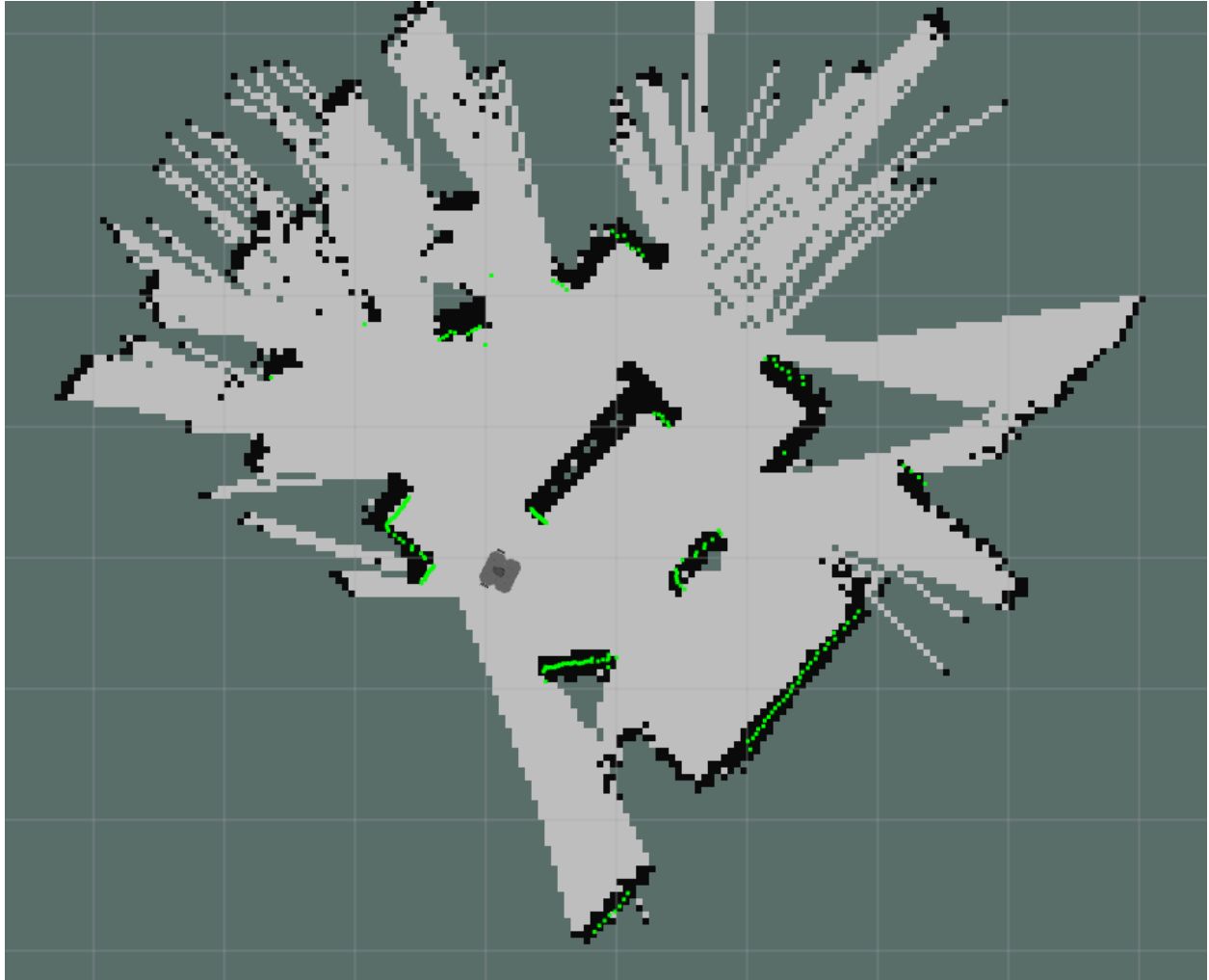


Figure 4: Real world map generated using GMapping algorithm in Myhal

#### Map of the Myhal Testing Environment

Our final map of the Myhal 570 testing environment shows partial coverage of the room's layout, including obstacles and walls detected by the LIDAR. While many walls appear correctly aligned, there are areas of the map with scattered or "spiky" artifacts, which primarily arise when the robot's estimated pose shifts abruptly. Despite these irregularities, much of the main region in front of the robot is mapped well, illustrating that scan matching generally succeeded in building a coherent layout.

### **Mapping Issues and Differences from Simulation**

One noticeable issue was the robot “jumping around” in RViz. These sudden pose shifts likely stem from the SLAM algorithm’s localization step, where Gmapping re-aligns the robot’s position whenever it detects a discrepancy between the predicted pose (from odometry) and the observed laser scans. In the real environment, factors such as uneven floors, wheel slippage, or slight LIDAR measurement errors can amplify these discrepancies, causing abrupt corrections in the robot’s reported pose. Additionally, reflective surfaces, people walking by, or other dynamic elements in Myhal can introduce noise that is less common or nonexistent in simulation. Unlike the highly controlled simulation environment, real-world conditions generate more unpredictable sensor readings, making localization and map construction more challenging and prone to these occasional jumps.